Quartiles, IQR, and Box Plots KEY

Name _____

Quartiles cut the data into **4 parts** with the same number of values. Each part has about **25%** of the data.

How to find the quartiles	**Five-number summary**
1. Put the data in order. 2. Find the **median**. It splits the data into a **lower half** and an **upper half**. 3. **Q1** = median of the lower half. 4. **Q3** = median of the upper half. **Odd number of values?** Leave the median out of both halves.	Minimum Q1 Median Q3 Maximum **IQR = Q3 − Q1** The IQR (interquartile range) is how spread out the **middle half** of the data is.

Example 1: odd number of values (11)

3, 4, 6, 7, 9, $\boxed{10}$, 12, 13, 15, 18, 20 Median = 10 (leave it out of the halves)

Lower half: 3, 4, $\boxed{6}$, 7, 9 $\rightarrow$ Q1 = 6 Upper half: 12, 13, $\boxed{15}$, 18, 20 $\rightarrow$ Q3 = 15

IQR = 15 − 6 = **9**

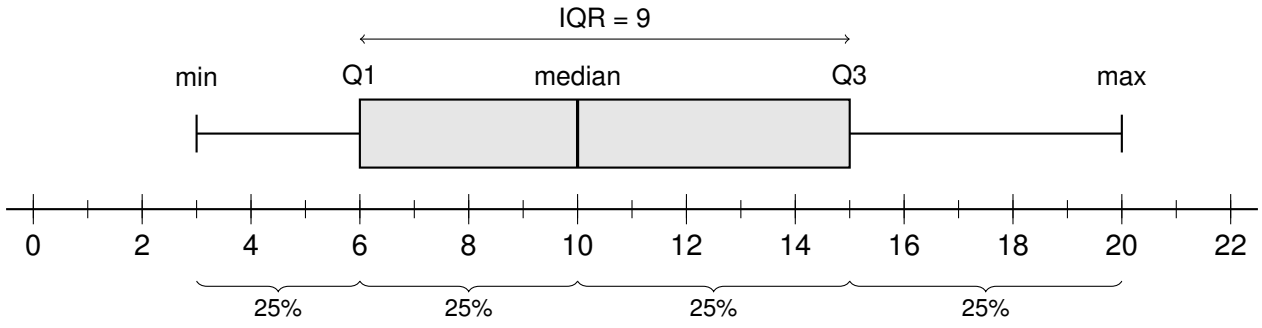
Example 2: even number of values (10)

1, 3, 4, 6, $\boxed{7}$, $\boxed{9}$, 10, 12, 14, 18 Median = (7 + 9) ÷ 2 = 8

Lower half: 1, 3, $\boxed{4}$, 6, 7 $\rightarrow$ Q1 = 4 Upper half: 9, 10, $\boxed{12}$, 14, 18 $\rightarrow$ Q3 = 12

IQR = 12 − 4 = **8**

A box plot draws the five-number summary. Here is Example 1.



Gotcha: The 4 parts have the same **number of values**, not the same width. A short part means the values are packed close together.

Work with a partner. Compare answers when you finish each problem.

1. Find the five-number summary and the IQR.

   4, 8, 1, 6, 9, 3, 7

   In order: 1, 3, 4, 6, 7, 8, 9. Min 1, Q1 3, Median 6, Q3 8, Max 9. IQR = 8 − 3 = 5

2. Find the five-number summary and the IQR.

   12, 15, 10, 20, 18, 11, 16, 14

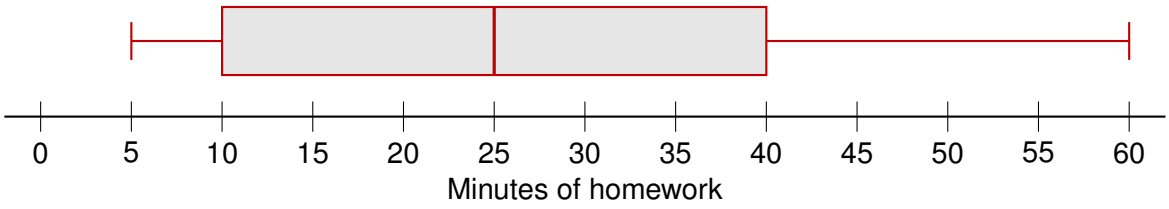
   In order: 10, 11, 12, 14, 15, 16, 18, 20. Median = (14 + 15) ÷ 2 = 14.5
   Lower half 10, 11, 12, 14: Q1 = 11.5. Upper half 15, 16, 18, 20: Q3 = 17.
   Min 10, Q1 11.5, Median 14.5, Q3 17, Max 20. IQR = 5.5

3. Eleven students recorded how many minutes they spent on homework last night.

   30, 10, 45, 5, 25, 60, 15, 30, 10, 40, 20

   (a) Find the five-number summary.
   5, 10, 10, 15, 20, 25, 30, 30, 40, 45, 60. Min 5, Q1 10, Median 25, Q3 40, Max 60

   (b) Make a box plot.



   (c) What is the IQR? What does it tell you?
   40 − 10 = 30 minutes. The middle half of the students spent between 10 and 40 minutes, a spread of 30 minutes.

4. Twelve students counted how many texts they sent in one hour.

13, 2, 20, 8, 30, 12, 5, 18, 10, 24, 8, 15

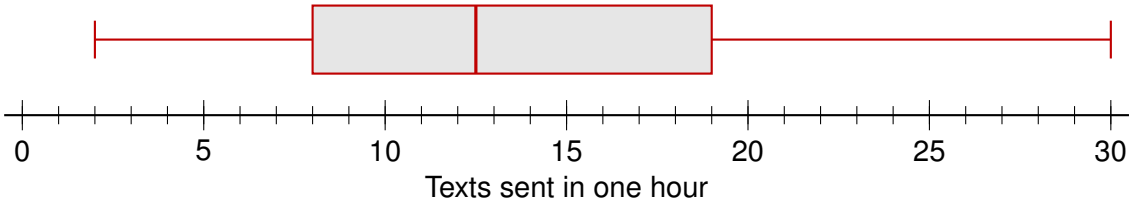
(a) Find the five-number summary.

2, 5, 8, 8, 10, $\boxed{12}$, $\boxed{13}$, 15, 18, 20, 24, 30. Median 12.5

Lower half 2, 5, 8, 8, 10, 12: Q1 = 8. Upper half 13, 15, 18, 20, 24, 30: Q3 = 19.

Min 2, Q1 8, Median 12.5, Q3 19, Max 30

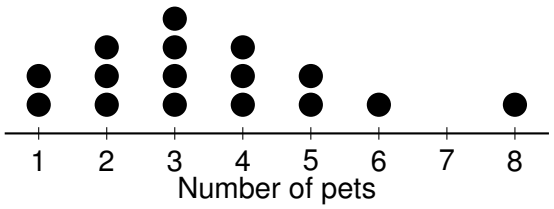
(b) Make a box plot.



(c) Which part of the box plot is the longest? What does that tell you about the data?

The right whisker (19 to 30). The top 25% of students are spread out the most.

5. This dot plot shows how many pets some students have.



(a) How many students are in the data? 16

(b) Find Q1, the median, and Q3.

1, 1, 2, 2, 2, 3, 3, $\boxed{3}$, $\boxed{3}$, 4, 4, 4, 5, 5, 6, 8. Median 3

Lower half: 1, 1, 2, 2, 2, 3, 3, 3 $\rightarrow$ Q1 = 2. Upper half: 3, 4, 4, 4, 5, 5, 6, 8 $\rightarrow$ Q3 = 4.5

(c) Find the IQR. $4.5 - 2 = 2.5$

6. Look at these two data sets.

Set A: 1, 2, 3, 4, 5, 6, 7

Set B: 1, 2, 3, 4, 5, 6, 70

(a) Find the IQR of each set.
Both: Q1 = 2, Q3 = 6, IQR = 4

(b) Find the range (max − min) of each set.
A: 7 − 1 = 6. B: 70 − 1 = 69

(c) One number changed. Which measure of spread changed a lot? Which did not change? Why?
The range changed a lot. The IQR did not change. The IQR only uses the middle half. One extreme value does not move it much. The median works the same way: one extreme value moves the mean a lot, but not the median.